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Studierichting: Informatica
Oefeningenreeks 4A nr. 12.9

$$\begin{aligned} & \int \frac{2^{x+2} + 5^{x-2}}{4^{x-1}} dx \\ &= \int \frac{2^{x+2}}{4^{x-1}} dx + \int \frac{5^{x-2}}{4^{x-1}} dx \\ &= \int \frac{2^{x+2}}{2^{2x-2}} dx + \int \frac{5^x \cdot 5^{-2}}{4^x \cdot 4^{-1}} dx \\ &= \int 2^{-x+4} dx + \frac{4}{5^2} \int \frac{5^x}{4^x} dx \\ &= 16 \int 2^{-x} dx + \frac{4}{25} \int \left(\frac{5}{4}\right)^x dx \\ &= \frac{16 \cdot 2^{-x}}{\ln 2} + \frac{4}{25} \cdot \left(\frac{1}{\ln \frac{5}{4}} \cdot \left(\frac{5}{4}\right)^x \right) + c \\ &= \frac{16 \cdot 2^{-x}}{\ln 2} + \frac{4 \left(\frac{5}{4}\right)^x}{25 \ln \frac{5}{4}} + c \end{aligned}$$

References